



Biochar Informatics

AI-Driven Biochar CO₂ Adsorption Intelligence Platform

Platform User Guide

Complete guide to using the Biochar Informatics AI tools and database

Version	v1.0 · May 2026
Language	English (EN)
Dataset	1,396 peer-reviewed experimental records
Platform	biocharassistant.com · ML Predictor Hub

1. Introduction

BiocharHub is an open-access, AI-powered web platform developed with instant access to a curated database of 1,396 peer-reviewed biochar CO₂ adsorption experiments, together with three machine-learning tools for rapid property prediction and material selection.

Platform Purpose

- Centralise global biochar experimental data in one searchable interface
- Deliver ML-powered CO₂ adsorption predictions from process conditions
- Guide material selection for carbon capture and sustainable agriculture
- Support reproducible, data-driven biochar research worldwide

1.1 Who Should Use This Guide?

- Researchers and graduate students working on biochar or carbon capture
- Environmental engineers evaluating adsorbent materials
- Policy analysts assessing carbon sequestration potential
- Educators and students learning about biochar science
- Developers integrating biochar data into larger workflows

1.2 Platform at a Glance

Feature	Details
Database Records	1,396 peer-reviewed experimental records
Experiments	59 unique experimental setups
ML Tools	CO ₂ Predictor · Property Estimator · Materials Advisor
Languages	English (EN) · Thai (TH)
Theme	Light & Dark mode
Access	Free, open-access — no login required

2. Getting Started

2.1 Accessing the Platform

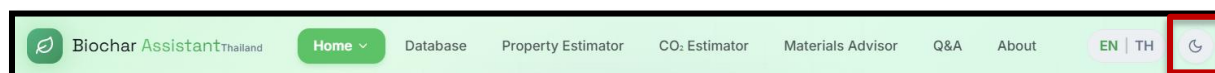
Open any modern web browser (Chrome, Firefox, Safari, or Edge) and navigate to the platform URL. The site is fully responsive and works on desktop, tablet, and mobile devices.

2.2 Language Selection



The platform supports English and Thai. The language toggle (EN | TH) is located in the top-right corner of every page. Clicking TH switches all navigation labels, descriptions, and UI text to Thai instantly — experimental data values remain in their original scientific notation.

2.3 Light / Dark Mode



A sun/moon icon next to the language toggle switches between Light and Dark colour themes. Your preference is remembered for the current session. The dark theme uses a deep navy-green palette optimised for extended reading sessions.

2.4 Navigation

The top navigation bar provides access to all platform sections:

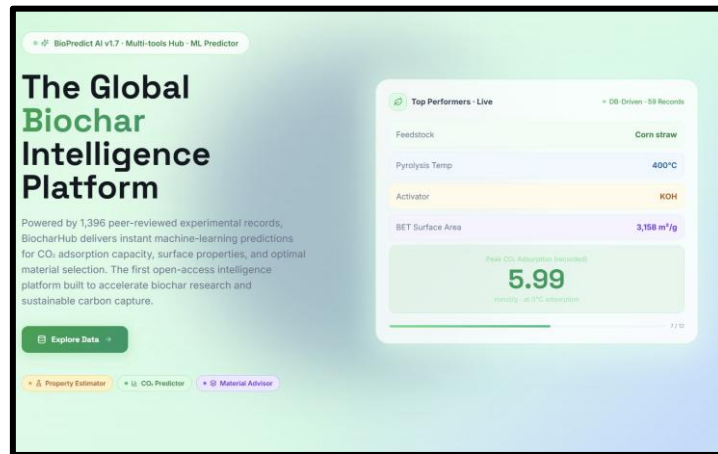


Nav Item	Destination	Purpose
Home	/	Platform overview, hero stats, tool launcher
Database	/database	Full experimental record browser
Property Estimator	/property-estimator	Predict biochar properties from process conditions
CO ₂ Estimator	/predictor	Predict CO ₂ adsorption capacity (mmol/g)
Materials Advisor	/advisor	AI-guided material & activator recommendations
About	/about	Research background, team, references

3. Home Page Overview

The Home page is the landing experience for first-time and returning visitors. It is divided into several sections, each serving a distinct informational or navigational purpose.

3.1 Hero Section



The hero section occupies the full viewport on load and consists of two columns:

- Left column — Platform title, extended description, the Explore Data button, and three tool shortcut pills (Property Estimator / CO₂ Predictor / Material Advisor). Each pill is a clickable hyperlink to the respective tool.
- Right column — A live 'Top Performers' card that cycles through the 12 highest-recorded CO₂ adsorption entries from the database every 3 seconds, showing feedstock, pyrolysis temperature, activator, BET surface area, and peak CO₂ uptake value.

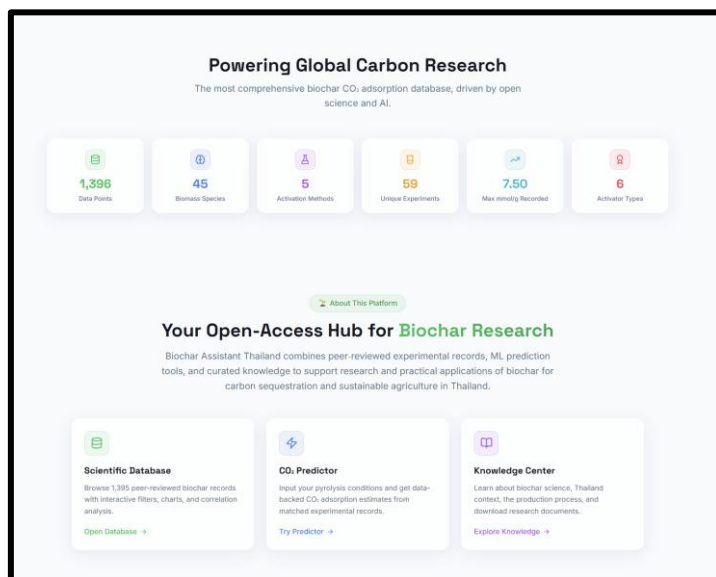
Tool Shortcut Pills

The three coloured pills below the Explore Data button are direct navigation links:

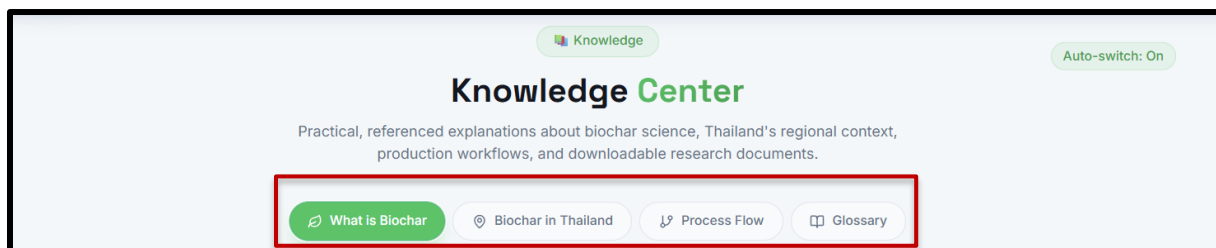
Pill	Colour	Navigates To
Property Estimator	Amber	/property-estimator
CO ₂ Predictor	Green	/predictor
Material Advisor	Violet	/advisor

3.2 Briefly guide about platform

After Hero session the next one is the tab that briefly highlight abilities of the platform which is total data points and 3 mains session: database, model prediction, and knowledge center

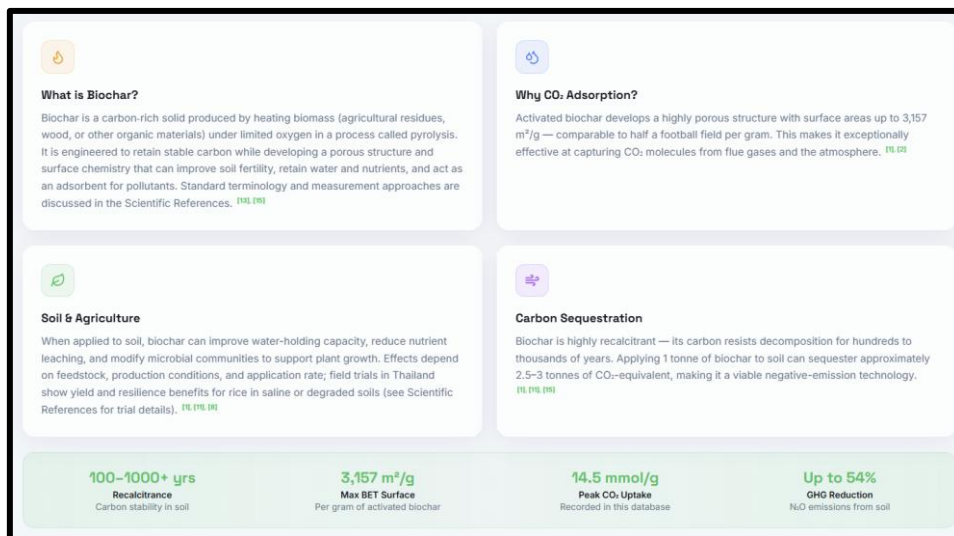


3.3 Knowledge center



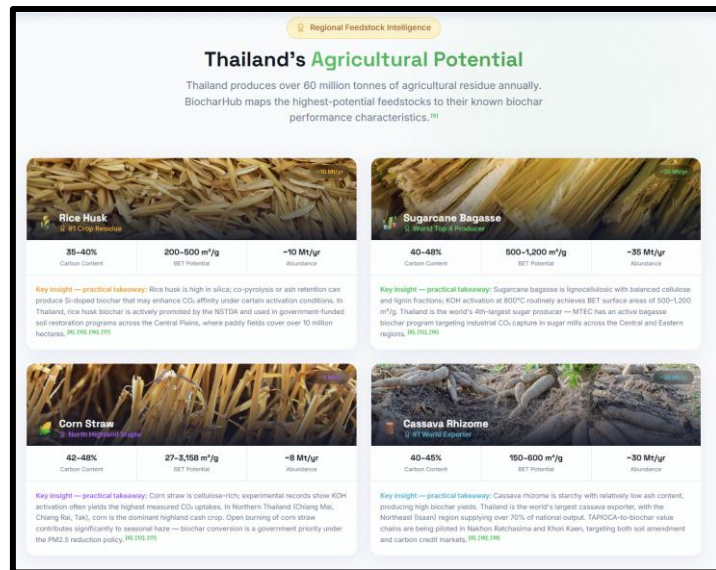
Knowledge center provided information about biochar for general users (untechnical user) it contain 4 different sub-session including:

1. What is Biochar



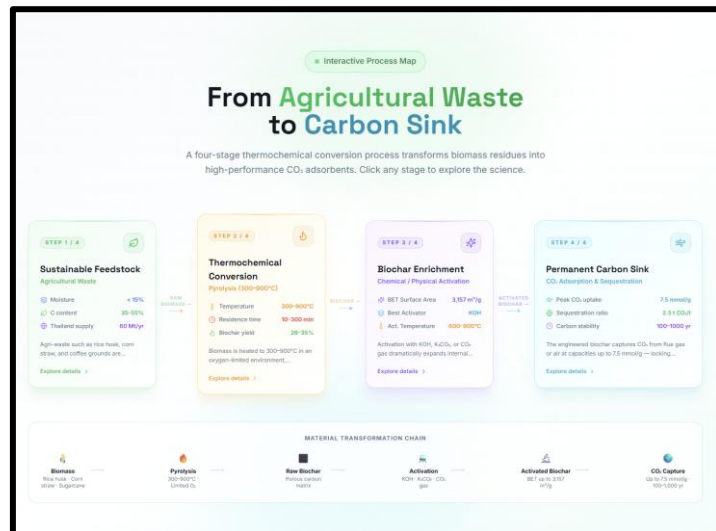
This will talk about basic knowledge about biochar and it applications

2. Biochar in Thailand



What is the most biomass was harvested in Thailand

3. Process Flow



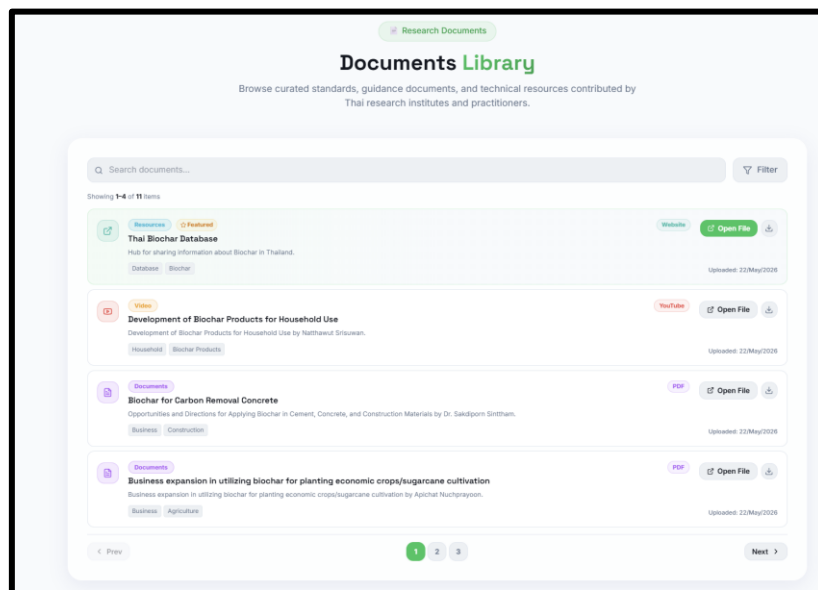
The work flow of Biomass to adsorption material especially for CO₂ adsorption application

4. Glossary

Biochar Glossary		
Plain-language definitions of technical terms, abbreviations, and units used in biochar research — click any card to expand.		
Search terms, abbreviations, or descriptions... AS General Characterization Activation Process Units		
Showing 28 of 28 terms		
Biochar Biochar Carbon-rich solid produced by heating biomass (crop residues, wood, etc.) at high temperature under very little... Click to read more >	Biomass Biomass / Feedstock The raw organic material used to produce biochar — e.g., sugarcane bagasse, rice husk, corn straw, pine sawdust. T... Click to read more >	Pyrolysis Pyrolysis The thermochemical process of heating organic material (300-900°C) in the absence of oxygen. This converts... Click to read more >
Adsorption Adsorption The process by which gas or liquid molecules (e.g., CO ₂) attach to and accumulate on the surface of a solid material... Click to read more >	Sequestration Carbon Sequestration Long-term removal of CO ₂ from the atmosphere and storage in a stable form. Biochar sequesters carbon by locking it in... Click to read more >	Recalcitrance Recalcitrance The resistance of biochar's carbon structure to biological or chemical breakdown. Highly recalcitrant biochar can persist... Click to read more >
BET Brunauer-Emmett-Teller (Surface Area) A standard laboratory technique that measures the total specific surface area of a porous material (in m ² /g) by... Click to read more >	Pore Volume Total Pore Volume The cumulative volume of all pores inside a biochar particle (cm ³ /g or m ³ /kg). A higher pore volume means more space... Click to read more >	Micropore Micropore (< 2 nm) Pores smaller than 2 nanometres in diameter. CO ₂ molecules (kinetic diameter ≈ 0.33 nm) are selectively trapped here... Click to read more >
Mesopore Mesopore (2-50 nm) Pores between 2 and 50 nm. They act as transport highways... Click to read more >	FTIR Fourier-Transform Infrared Spectroscopy An analytical technique that identifies chemical functional... Click to read more >	XRD X-ray Diffraction A technique that analyses the crystalline structure of biochar... Click to read more >

The keyword that normally found in biochar research and application area

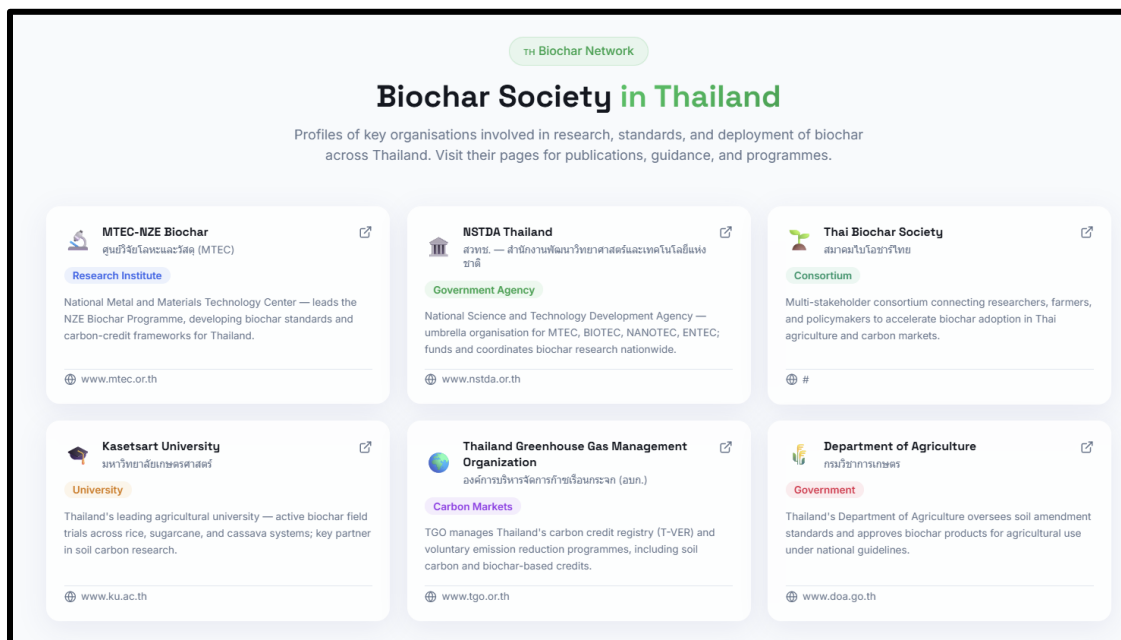
3.3 Documents Library

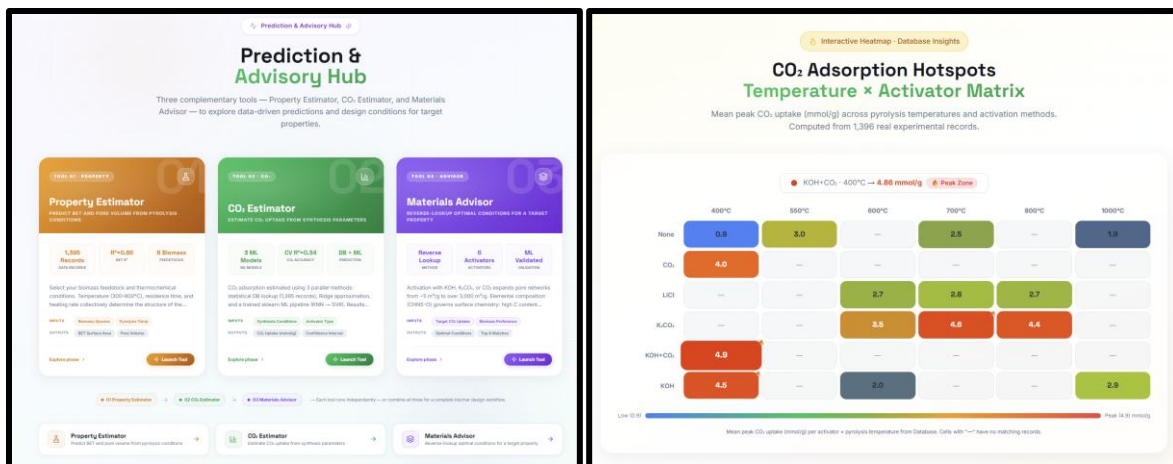


Information hub that update the news about biochar in Thailand

3.4 Other sub-session in Home page

Scrolling past the document's library reveals additional sections: biochar society in Thailand, the Prediction & Advisory Hub explaining the prediction workflow, a Temperature × Activator heatmap visualization, and Q&A.





[Help Centre · FAQ](#)

Questions & Answers

Find answers to common questions about the platform, ML tools, and the science behind biochar CO₂ adsorption research.

General (4)

CO₂ Predictor (4)

Property Estimator (3)

Materials Advisor (3)

Science (3)

General · 4 questions

What is BiocharHub?

Is BiocharHub free to use?

Who should use this platform?

Where does the data come from?

Still need help?

Reach the research team via the About page.

[Visit About page →](#)

View All Q&A

29 questions · 6 categories

[Open Help Centre →](#)

Ready to get started?

Run a prediction or explore the database.

[Predict](#)
[Database](#)

3.5 Reference Sections



Scientific References

All factual claims on this page are supported by peer-reviewed literature or primary data sources.

[1] Lehmann, J., Gaunt, J., & Rondon, M. (2006). **Bio-char sequestration in terrestrial ecosystems – a review.** *Mitigation and Adaptation Strategies for Global Change*, 11, 403–427.
doi.org/10.1007/s11027-005-9006-5

[2] Woolf, D., Amonette, J. E., Street-Perrott, F. A., Lehmann, J., & Joseph, S. (2010). **Sustainable biochar to mitigate global climate change.** *Nature Communications*, 1, 56.
doi.org/10.1038/ncomms1053

[3] IPCC (2022). **Mitigation of Climate Change. Contribution of Working Group III to the Sixth Assessment Report (AR6).** *Intergovernmental Panel on Climate Change*, Cambridge University Press.
www.ipcc.ch/report/ar6/wg3/

[4] European Biochar Certificate (EBC) (2023). **Guidelines for a Sustainable Production of Biochar. EBC**, Version 10.1. Arbaz, Switzerland.
www.european-biochar.org/en/ebc-guidelines

[5] Huggins, T. M., Haeger, A., Biffinger, J. C., & Ren, Z. J. (2016). **Granular biochar compared with activated carbon for wastewater treatment and resource recovery.** *Water Research*, 94, 225–232.
doi.org/10.1016/j.watres.2016.02.059

[6] Shen, W., Zhang, S., He, Y., Fang, J., & Tang, Y. (2011). **Hierarchical pore structure of activated carbon derived from corn stalks.** *Journal of Materials Science*, 46, 464–469.
doi.org/10.1016/j.jmatsci.2016.07.069

Database-derived statistics (marked [12]) are calculated directly from the BiocharHub Database.xlsx experimental records. All other figures cite published peer-reviewed sources. Last reviewed: April 2026.

References and scientific references list that used in this platform were concluded in here.

4. Database Explorer

The Database page provides access to all 1,396 experimental records collected from peer-reviewed literature. Each record captures the full experimental chain from biomass feedstock through production and activation to final CO₂ adsorption measurement.

4.1 Record Structure

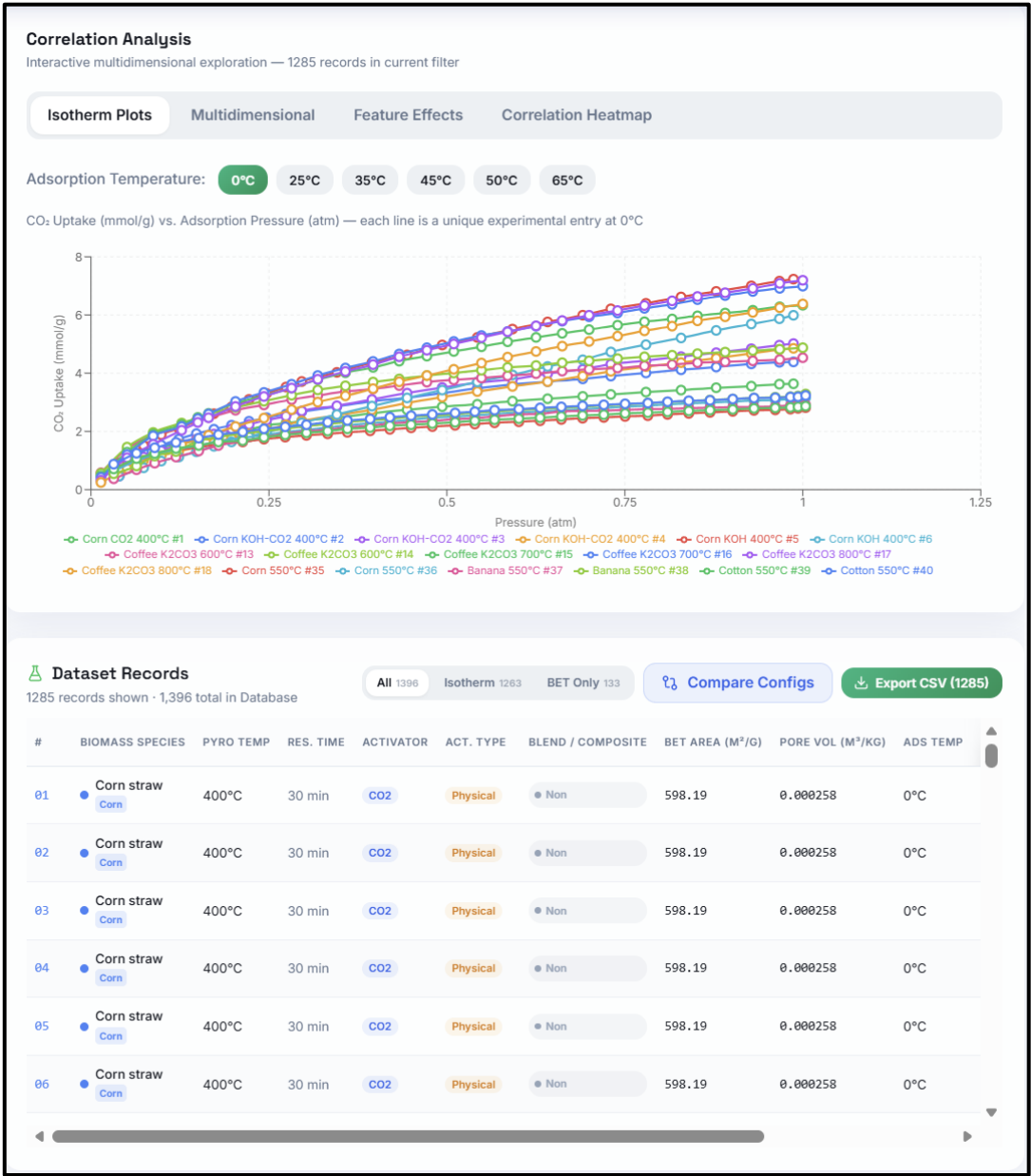
Field	Description
Biomass / Feedstock	Raw material source (e.g., rice husk, corn straw, coffee grounds)
Pyrolysis Temperature	Temperature (°C) of the carbonisation step
Activation Agent	Chemical or physical activator applied post-pyrolysis
Activation Temperature	Temperature (°C) of the activation step
BET Surface Area	Measured pore surface area (m ² /g)
Pore Volume	Total pore volume (cm ³ /g)
Average Pore Size	Mean pore diameter (nm)
Adsorption Temperature	CO ₂ measurement temperature (°C)
CO ₂ Uptake	Measured CO ₂ adsorption capacity (mmol/g)
Reference	DOI / citation to source publication

4.2 Filtering & Sorting

The image displays the filtering interface of the Database Explorer. It features several panels for narrowing down records:

- Filters Panel:** Includes a search box for "Search biomass, activator" and buttons for "All", "Isotherm", and "BET Only".
- BIOMASS FEEDSTOCK:** A list of feedstock types with checkboxes and counts: Algae, Bamboo, Banana (2), Coconut (4), Coffee, Corn (3), Cotton (2), Durian, Palm (3), Pine (5), Pomelo, Rice (2), Sugarcane (3), Wheat (2), Wood (9), and Other (5).
- ACTIVATOR:** A list of activators with checkboxes: CO₂, K₂CO₃, KOH, KOH-CO₂, LiCl, and Non.
- ACTIVATION TYPE:** A list of activation types with checkboxes: Chemical, Combined, Non, and Physical.
- BLEND / COMPOSITE:** A list of blend/composite types with checkboxes: 0.5PKBC, 0.5TKBC, 20PKBC, 20TKBC, and Non.
- ADSORPTION TEMP (°C):** A list of adsorption temperatures with checkboxes: 25, 35, 45, 50, and 65.
- BET SURFACE AREA:** A range filter from 0 m²/g to 3,200 m²/g, with a slider and input fields for Min and Max.
- PORE VOLUME (×10⁻³):** A range filter from 0 to 1,600, with a slider and input fields for Min and Max.
- CO₂ UPTAKE:** A range filter from 0 mmol/g to 8 mmol/g, with a slider and input fields for Min and Max.
- PYROLYSIS TEMP:** A range filter from 300°C to 900°C, with a slider and input fields for Min and Max.

- Use the filter panel to narrow records by feedstock type, activation agent, temperature range, or CO₂ uptake range
- Click any column header to sort ascending or descending
- Use the search box for free-text search across all fields
- Export filtered results to CSV for offline analysis



5. Property Estimator

The Property Estimator predicts physical and chemical properties of biochar given its production conditions. This tool is useful when you have a planned production process and want to estimate the resulting material characteristics before running experiments.

5.1 Predicted Properties

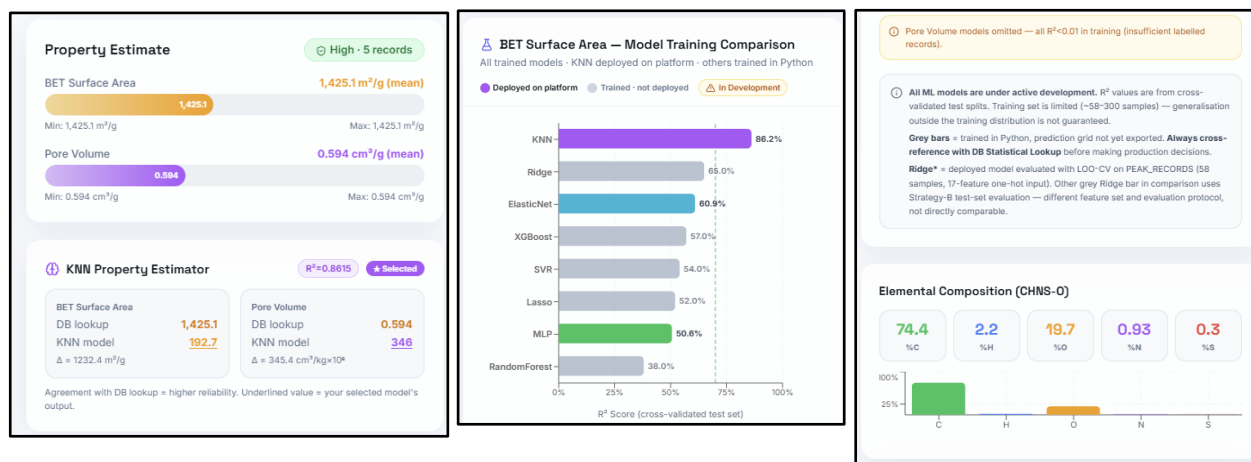
Property	Unit	What It Represents
BET Surface Area	m ² /g	Pore surface area — key indicator of adsorption capacity
Pore Volume	cm ³ /g	Total volume of pores available for gas uptake
Average Pore Size	nm	Average diameter of pores (micro/mesopore distribution)
Carbon Content	%	Elemental carbon fraction of the biochar mass
Yield	%	Mass fraction of biochar produced from the feedstock

5.2 How to Use

The screenshot shows the Property Estimator interface. A red box highlights the input fields: Biomass Species (Corn straw), Pyrolysis Temperature (600 °C), Residence Time (60 min), Activation Method (KOH), and the ML Model section. Red arrows point from these fields to labels: Biomass, Pyrolysis condition, Activator, and ML model. The ML Model section shows three options: KNN Regressor (selected), ElasticNet, and MLP Neural Network. The KNN Regressor is described as a K-Nearest Neighbours model trained on pyrolysis conditions, with a best-validated R² of 0.8615. The ElasticNet is described as an Elastic Net regularisation (L1+L2) model, with a best-validated R² of 0.6091. The MLP Neural Network is described as a Multi-Layer Perceptron model, with a best-validated R² of 0.506. The interface also includes a status bar at the top indicating 'Inputs: Pyrolysis conditions matched against Database records' and a button at the bottom labeled 'Estimate Properties'.

- Select your feedstock type and pyrolysis conditions
- Add activation details if applicable
- Click 'Estimate Properties' to run the ML model

- Review the predicted property table and compare against known experimental benchmarks



5.3 Save/Export the prediction Results

BiocharInformaticsThailand – Property Estimation Report The Global Biochar Intelligence Platform	
Generated : 2 June 2026 at 10:09 Platform : BiocharInformaticsThailand Property Estimator V.1.0 Source : 44-record peer-reviewed BET database	
SYNTHESIS PARAMETERS	
Biomass Species	: Corn straw
Pyrolysis Temperature	: 600 °C
Residence Time	: 60 min
Activation Method	: KOH
ESTIMATION CONFIDENCE	
Confidence Level	: High
Data Points Used	: 5 matching records
STRUCTURAL PROPERTIES – DATABASE LOOKUP	
BET Surface Area	
Mean	: 1,425.1 m ² /g
Min	: 1,425.1 m ² /g
Max	: 1,425.1 m ² /g
Pore Volume	
Mean	: 0.594 cm ³ /g
Min	: 0.594 cm ³ /g
Max	: 0.594 cm ³ /g
ELEMENTAL COMPOSITION (CHNS-O)	
Carbon (C)	: 74.4 %
Hydrogen (H)	: 2.2 %
Oxygen (O)	: 19.7 %
Nitrogen (N)	: 0.93 %
Sulphur (S)	: 0.3 %

After finishing the prediction, user can export the prediction results as .txt file

6. CO₂ Predictor (CO₂ Estimator)

The CO₂ Predictor is the core ML tool of the platform. It accepts biochar production parameters and returns an estimated CO₂ adsorption capacity in mmol/g, derived from matched experimental records using a k-nearest-neighbour similarity search across the 1,396-record database.

6.1 Input Parameters

Data Source: Expert guidance based on 1,396 peer-reviewed experimental records

1 Select Biomass

Biomass / Feedstock Type

Corn straw

8 species with confirmed experimental records

2 Pyrolysis Conditions

Pyrolysis Temperature

600 °C

350°C 900°C

Residence Time

60 min

10 min 300 min

Heating Rate

10 °C/min

1°C/min 20°C/min

3 Activation Method

Chemical / Physical Activator

None (No Activation)

4 Chemical Blend — optional New - Limited data

Mix biochar with a chemical composite (PKBC / TKBC). Enter the chemical name and weight %. Effect is interpolated from experimental records.

No blend (pure biochar)

5 ML Model — choose your ML predictor

DB Statistical Lookup always shows. Select which ML model to highlight as your primary prediction alongside it.

Ridge Regression Linear - Always Available R²=0.4455

Trained Ridge (α=0.01) on 58 PEAK_RECORDS, 17-feature one-hot input: biomass + activator + pyrolysis conditions.

Trained on real data - always available - no grid needed

Linear — cannot capture non-linear activation interactions

KNN → SVR Pipeline Pipeline - Best CO₂ R² R²=0.3427

Two-stage: KNN estimates BET surface area, SVR predicts CO₂ from BET features. Best cross-validated CO₂ model.

Best test-set R² for CO₂ - also outputs BET surface area

Pre-computed lookup grid (1,728 pts) - conditions outside grid interpolated

Stacking Ensemble Ensemble - Deployed R²=0.247

Meta-learner combining SVR + KNN + RF + XGBoost as base models with SVR as the meta-learner. Strategy-B CV R²=0.25.

Deployed - ensemble reduces variance vs single models

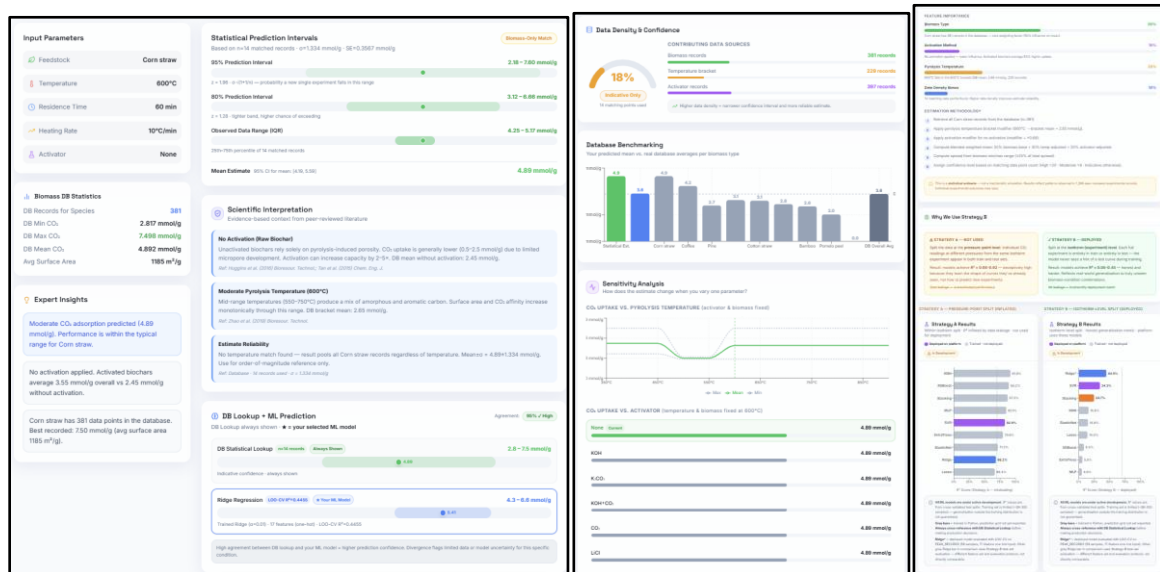
Lower R² than SVR on this dataset - largest prediction grid

Generate Expert Guidance

BioPredict AI V.1.0 · Statistical + ML Pipeline · Based on Historical Research Data

Parameter	Unit / Options	Description
Feedstock (Biomass)	Text / dropdown	Raw biomass material used to produce the biochar
Pyrolysis Temperature	°C (300–1200)	Temperature at which pyrolysis was performed
Activation Method	None / KOH / K ₂ CO ₃ / CO ₂ / KOH+CO ₂ / LiCl	Chemical or physical activation agent applied
Activation Temperature	°C (optional)	Temperature used during the activation step
BET Surface Area	m ² /g (optional)	Brunauer–Emmett–Teller surface area measurement
Adsorption Temperature	°C (0, 25, 50)	Target temperature for CO ₂ adsorption measurement

6.2 Reading the Results



After submitting, the predictor displays:

- Predicted CO₂ Adsorption — the estimated uptake value in mmol/g
- Confidence range — the spread of matched experimental records
- Matched records — the closest experimental entries from the database used to generate the estimate
- Similarity score — how closely the input matches the retrieved records

Interpreting mmol/g Values

- < 2.0 mmol/g → Low adsorption capacity
- 2.0 – 4.0 mmol/g → Moderate capacity (typical for unactivated biochar)
- 4.0 – 6.0 mmol/g → High capacity (activated biochar)
- > 6.0 mmol/g → Exceptional capacity (top-performing materials)

6.3 Tips for Best Results

- Provide BET Surface Area if available — it significantly improves prediction accuracy
- KOH activation typically yields the highest CO₂ uptake across the dataset
- Pyrolysis temperatures of 600–800°C tend to produce the best-performing biochar
- Adsorption at 0°C yields higher uptake values than 25°C or 50°C

6.4 Save/Export the prediction Results

```

BiocharInformaticsThailand · BioPredict AI V.1.0 - Estimation Report
Generated: 6/2/2026, 10:08:59 AM
Total Database Records: 1,396 peer-reviewed experimental data points

--- INPUT PARAMETERS ---
Biomass / Feedstock : Corn straw
Pyrolysis Temperature: 600°C
Residence Time      : 60 min
Heating Rate        : 10°C/min
Activator           : None

--- ESTIMATION RESULTS ---
Predicted Min CO2 Uptake : 2.82 mmol/g
Predicted Mean CO2 Uptake: 4.89 mmol/g ← Recommended Target
Predicted Max CO2 Uptake : 7.5 mmol/g
Confidence Level     : Indicative
Matching Data Points Used: 14

--- BIOMASS DATABASE STATISTICS ---
Biomass : Corn straw
DB Records : 381
DB Min CO2 : 2.817 mmol/g
DB Max CO2 : 7.498 mmol/g
DB Mean CO2 : 4.892 mmol/g
Avg Surface Area: 1185 m2/g

--- ACTIVATOR STATISTICS ---
Activator: None
DB Mean : 2.45 mmol/g
DB Count : 397 records

--- EXPERT INSIGHTS ---
[NEUTRAL] Moderate CO2 adsorption predicted (4.89 mmol/g). Performance is within the typical range for Corn straw.
[INFO] No activation applied. Activated biochars average 3.55 mmol/g overall vs 2.45 mmol/g without activation.
[INFO] Corn straw has 381 data points in the database. Best recorded: 7.50 mmol/g (avg surface area 1185 m2/g).

--- DISCLAIMER ---
Results are statistical estimates derived from aggregated
peer-reviewed research data. Raw individual records are
never exposed. Individual experimental outcomes may vary.
This report is for research planning purposes only.

BiocharInformaticsThailand · biocharinformaticsthailand.ai · BioPredict AI V.1.0

```

After finishing the prediction, user can export the prediction results as .txt file

7. Materials Advisor

The Materials Advisor is a goal-oriented recommendation tool. Instead of inputting process conditions, you describe your performance target (e.g., maximise CO₂ adsorption at 25°C) and the Advisor recommends optimal feedstock, pyrolysis temperature, and activation strategy based on the full database.

7.1 Recommendation Workflow

Define Your Target
Set your desired CO₂ uptake and we'll find the production parameters that achieve it in the research database.

Target CO₂ Uptake
 Slider: 0.1 mmol/g (min in DB) to 7.5 mmol/g (max in DB)
 Value: 4.5 mmol/g
 Target: 4.5 mmol/g - Above average (strong activation needed)

Acceptable Tolerance (±mmol/g)
 Slider: ±0.75
 Narrow = precise match · Wide = more options

Secondary Objective (optional)
 None
 ⚡ Minimize Pyrolysis Temperature (Energy Saving)
 🔥 Minimize Chemical Activator (Material Cost Saving)
 ⌚ Minimize Residence Time (Process Speed)
 📦 Maximize Pore Volume (Adsorption Capacity)

Tertiary Objective (optional)
 None
 ⚡ Minimize Pyrolysis Temperature (Energy Saving)
 🔥 Minimize Chemical Activator (Material Cost Saving)
 ⌚ Minimize Residence Time (Process Speed)
 📦 Maximize Pore Volume (Adsorption Capacity)

Biomass Preference (optional)
 Any Species
 Algae, Bamboo, Banana, Coconut, Coffee, Corn, Cotton, Durian, Palm, Pine, Pomelo, Rice, Sugarcane, Wheat, Wood, Other

Find Optimal Parameters

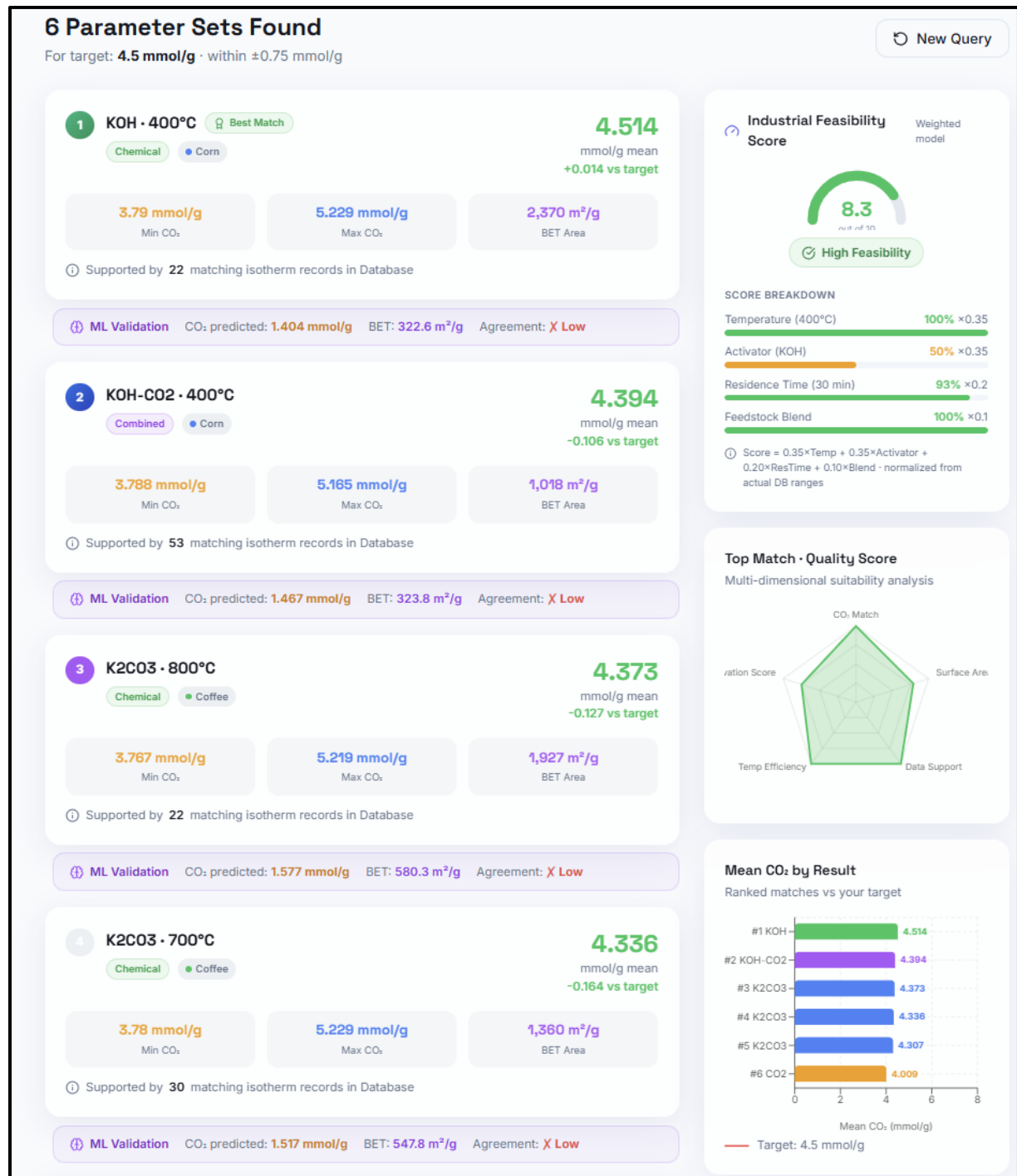
- Step 1 — Set your objective (maximise CO₂ uptake, maximise surface area, target a cost-effective production route)
- Step 2 — Apply optional constraints (available feedstocks, temperature limits, no chemical activation)
- Step 3 — Review ranked recommendations with predicted performance scores
- Step 4 — Explore the matched experimental records supporting each recommendation

7.2 Common Use Cases

Example Objectives

→ 'Find the best biochar for CO₂ capture at room temperature (25°C)'

- 'Suggest a biochar from agricultural waste with no chemical activation'
- 'What feedstock gives the highest BET surface area with KOH activation?'
- 'Recommend a low-cost production route for ≥ 4 mmol/g CO₂ uptake'



Scientific Interpretation

RECOMMENDED ACTIVATOR

KOH

Chemical activation · highest match score in database

OPTIMAL PYROLYSIS TEMP

400°C

Avg BET surface area: 2,370 m²/g at this condition

EXPECTED OUTCOME

3.79–5.229 mmol/g

Range from 22 peer-reviewed records

7.3 Save/Export the prediction Results

```
|| BiocharInformaticsThailand – Materials Advisor Report ||
|| The Global Biochar Intelligence Platform ||

Generated : 2 June 2026 at 10:10
Platform : BiocharInformaticsThailand Materials Advisor V.1.0
Method : Reverse-lookup · Filter → Rank → Cluster

QUERY PARAMETERS

Target CO2 Uptake : 4.5 mmol/g
Acceptable Tolerance : ±0.75 mmol/g
Biomass Preference : Any species
Secondary Objective : None
Tertiary Objective : None
Result Mode : Exact match (within tolerance)
Total Results Found : 6

RECOMMENDATION #1 ★ BEST MATCH|

Activation Method : KOH
Activation Type : Chemical
Pyrolysis Temperature: 400 °C
Mean CO2 Uptake : 4.514 mmol/g
CO2 Range : 3.79 – 5.229 mmol/g
vs Target : +0.014 mmol/g
Avg BET Surface Area : 2,370 m2/g
Supporting Records : 22 peer-reviewed experiments
Biomass Species : Corn straw

RECOMMENDATION #2

Activation Method : KOH-CO2
Activation Type : Combined
Pyrolysis Temperature: 400 °C
Mean CO2 Uptake : 4.394 mmol/g
CO2 Range : 3.788 – 5.165 mmol/g
vs Target : -0.106 mmol/g
Avg BET Surface Area : 1,018 m2/g
Supporting Records : 53 peer-reviewed experiments
Biomass Species : Corn straw
```

After finishing the prediction, user can export the prediction results as .txt file

9. Q&A

Knowledge Centre · Help

Q&A / Help Centre

Find answers to common questions about the platform, ML tools, and the science behind biochar CO₂ adsorption.

General
CO₂ Predictor
Property Estimator
Materials Advisor
Database
Science

CATEGORIES

- General 6
- CO₂ Predictor 6
- Property Estimator 4
- Materials Advisor 4
- Database 4
- Science 5

Science · 5 questions

- What is biochar and how is it made?
- Why does pyrolysis temperature affect CO₂ adsorption so much?
- How does chemical activation improve CO₂ adsorption?
- What does mmol/g actually mean in physical terms?
- What are micropores and why do they matter for CO₂ capture?

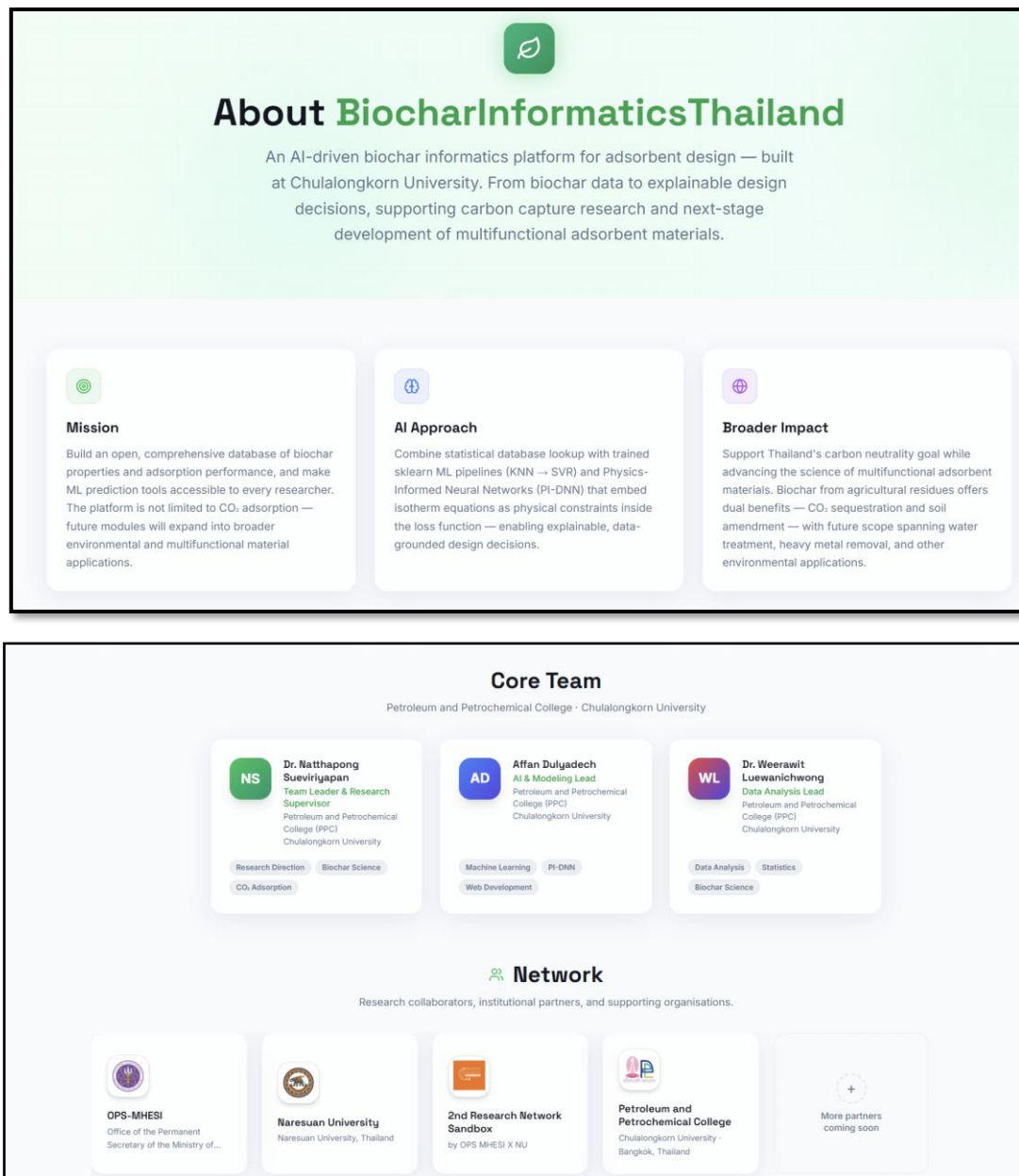
Still need help?
Reach the research team via the About page.
[Visit About page →](#)

Ready to get started?
Explore the database or run a prediction.

Explore Database
Try the Tools

A list of Frequently Asked Questions categorized into 6 group: General, CO₂ Predictor, Property Estimator, Materials Advisor, Database, and Science question. This will help user more understand the content in this platform and basic information about biochar for adsorption applications.

9. About



Content

- Purpose: informational overview for BiocharHub with project mission, team, timeline, tech, and CTAs.
- Mission/AI/Impact: three feature cards explaining goals and modeling approach.
- Team: core team and network.
- Timeline: project milestones (including a data-driven milestone showing published datapoints and peak value).
- Tech stack: short list of key tools (React/Vite, scikit-learn, PyTorch, Tailwind, etc.).